LegalEngine — Technical Project Guide

Overview

LegalEngine is a **Chekin-branded compliance hub** that helps vacation rental owners and property managers understand and navigate the legal requirements for registering their properties across Spain and Europe. It combines a real-time AI chatbot powered by Claude, downloadable PDF compliance guides, and lead capture for legal services.

Live URL: `http://localhost:3000` (Node.js + Express)

Project Structure

```
LegalEngine/
├── index.html              ← Single-page frontend (HTML/CSS/JS, no framework)
├── server.js                ← Express backend (API + static file server)
├── package.json             ← Node.js dependencies and scripts
├── .env                     ← Environment variables (ANTHROPIC_API_KEY)
├── .env.example             ← Template for environment variables
├── background.jpg           ← WhatsApp-style doodle chat background
├── Guias/                   ← Source PDF guides (10 regional guides)
│   ├── GUÍA PARA ANDALUCÍA.pdf
│   ├── GUÍA PARA CATALUÑA.pdf
│   ├── GUÍA PARA LA COMUNIDAD VALENCIANA.pdf
│   ├── GUÍA PARA LAS ISLAS BALEARES.pdf
│   ├── GUÍA ACTUALIZADA TFE-GC-LTE-FTE.pdf
│   ├── GUÍA ACTUALIZADA LA GOMERA-EL HIERRO-LA PALMA.pdf
│   ├── GUÍA PARA MADRID.pdf
│   ├── GUÍA GALICIA.pdf
│   ├── GUÍA CANTABRIA.pdf
│   └── GUÍA MURCIA.pdf
├── Guideformat.pdf          ← Visual style template for PDF guide generation
├── scripts/
│   └── extract-guides.js    ← PDF-to-Markdown extraction script
├── knowledge/               ← Extracted guide content (generated by extract script)
│   ├── _index.json          ← Manifest: slug, name, aliases, type, source file
│   ├── andalucia.md         ← Regional guides (10 files)
│   ├── cataluna.md
│   ├── valencia.md
│   ├── baleares.md
│   ├── canarias-main.md
│   ├── canarias-green.md
│   ├── madrid.md
│   ├── galicia.md
│   ├── cantabria.md
│   ├── murcia.md
│   ├── city-tax.md          ← Thematic guides (3 files)
│   ├── tourist-license.md
│   └── identity-verification.md
└── PROJECT_GUIDE.md         ← This file
```

Architecture

Frontend (`index.html`)

A single-page application with no build step — pure HTML, CSS, and vanilla JavaScript. Key sections:

Section	Description
Header	Chekin-branded mega-menu navigation (Features, Solutions, Resources, Pricing, Countries) with dropdowns, language selector (EN/ES), and CTAs
Hero	Search bar with typewriter placeholder animation. Transitions to chat mode on first question
Chat Interface	WhatsApp-style chat with SSE streaming, follow-up suggestions, "New topic" button
Region Cards	Grid of Spanish regions + European countries with guide download capability
Lawyer CTA	Lead capture form for connecting users with specialized lawyers
FAQ	Accordion-style frequently asked questions

Key Frontend Features:

- **SSE Streaming**: Real-time token-by-token rendering from Claude API
- **Region Detection**: Client-side `detectRegionSlug()` maps 50+ city/province names to guide slugs
- **Thematic Detection**: `isThematicQuestion()` identifies topic-based questions (invoicing, taxes, insurance, etc.) to skip the region picker
- **PDF Generation**: Uses jsPDF (v2.5.1) to draw PDFs directly — cover page, step pages, checklist — no html2canvas dependency
- **Bilingual Guides**: PDF content available in EN/ES, selected based on `data-lang` attribute
- **Lead Capture**: Two modal forms — one for lawyer referrals, one for guide downloads — with property type toggles and country/region pre-fill

Backend (`server.js`)

Express server handling static files and the AI API.

Endpoint	Method	Description
`/api/ask`	POST	Main chat endpoint. Accepts `{question, region?, history[], stream?}`. Returns SSE stream or JSON
`/api/regions`	GET	Returns list of available regional guides (excludes thematic guides)
`/health`	GET	Health check endpoint

Core Backend Logic:

1. **Knowledge Loading**: On startup, reads `knowledge/_index.json` and loads all `.md` guide files into memory (~30KB total)

2. **Guide Resolution** (`resolveGuides()`):
    - Checks explicit `region` parameter first
    - Scans question text against guide aliases and `CITY_TO_REGION` map
    - Fires `THEMATIC_TRIGGERS` regex patterns (city-tax, tourist-license, identity-verification)
    - Checks `EUROPEAN_LOCATIONS` array for non-Spanish locations
    - If no match found, returns empty (lets Claude use its own knowledge)
3. **Rate Limiting**: Simple in-memory rate limiter (15 requests/minute/IP)

AI Integration

Model & SDK

- **Model**: Claude Sonnet (`claude-sonnet-4-20250514` by default, configurable via `CLAUDE_MODEL` env var)
- **SDK**: `@anthropic-ai/sdk` (official Anthropic Node.js SDK)
- **Streaming**: Server-Sent Events (SSE) for real-time response rendering

How It Works

```
User question
     │
     ▼
resolveGuides() ──► Matches regional + thematic guides
     │
     ▼
Build messages array:
  [0] Guide context + first user message
  [1..n] Conversation history
  [n+1] Current question
     │
     ▼
Claude API (streaming) ──► System prompt defines expert persona
     │
     ▼
SSE chunks ──► Frontend renders token-by-token
     │
     ▼
Extract <follow-up> tags ──► Render as clickable suggestion buttons
```

System Prompt Design

The system prompt establishes Claude as the **Chekin Compliance Assistant** with these rules:

1. **Primary source**: Answer from guide content when available
2. **Supplemental knowledge**: When guides don't cover a topic, Claude can use its own knowledge (flagged as such)
3. **Language lock**: Always respond in the language of the user's first message — never switch mid-conversation
4. **HTML formatting**: Responses use `<strong>` , `<ul>/<li>` , `<ol>/<li>` , `<p>` for rich chat rendering

5. **Follow-up generation**: Every response ends with `<follow-up><q>...</q></follow-up>` tags for suggested next questions
6. **Context continuity**: Short follow-ups (city names, "yes"/"no") are treated as topic continuations, not new queries

Guide Context Injection

Relevant guides are prepended to the first user message as XML:

```
<guide region="Andalucia">
[Full markdown content of the guide]
</guide>

Question: How do I register my VUT in Malaga?
```

For conversations with history, the guide context is injected into the reconstructed first message, and subsequent messages are appended as-is. This preserves conversation flow while keeping the guide context available.

Thematic vs Regional Routing

The system distinguishes between two types of questions:

Type	Example	Behavior
Regional	"How do I register in Barcelona?"	Resolves to `cataluna` guide, sends as context
Thematic	"What is the city tax in Rome?"	Fires `city-tax` thematic trigger, sends thematic guide
General	"Do I need electronic invoicing?"	No guide match → Claude answers from own knowledge
Follow-up	"What about Huelva?" (after invoicing question)	Resolves `andalucia` guide from combined conversation text, maintains topic

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PDF Guide Generation

Approach

Uses **jsPDF** (v2.5.1) to draw PDFs programmatically — no HTML rendering or screen capture. Each PDF is constructed page-by-page using the jsPDF drawing API.

PDF Structure

1. **Cover Page**: Dark navy (#0a0a2e) background, badge, title, region name in purple, intro text, version metadata, Chekin logo
2. **Step Pages** (6 per guide): Purple step number badge, title, subtitle, divider, content blocks (paragraphs, bullet lists, stat cards, subsections with letter circles)
3. **Checklist Page**: Green check badge, numbered checklist items with purple checkboxes, CTA callout

Content Block Types

Block Type	Rendering
`p`	Paragraph text with automatic line wrapping
`li`	Bullet point with purple dot
`h4`	Bold subheading
`stats`	Horizontal row of stat cards (rounded rectangles with large number + label)
`sub`	Subsection card with letter circle (A, B, C), title, and description

Bilingual Support

Guide data is structured as `guideData[slug].en` and `guideData[slug].es`. The `getSiteLang()` function reads `document.documentElement.getAttribute('data-lang')` to determine which version to render. The language selector in the header updates this attribute.

Lead Capture

Two lead capture forms capture contact information:

1. Lawyer Referral Form (`#lawyerOverlay`)

- **Fields**: Country, Region, Email, Phone, Name, Company, Property type (toggle: Hotel/Apartment/Vacation Rental/Other), Number of units
- **Trigger**: "Talk to a lawyer" hero button + "Connect with a lawyer" CTA section
- **Pre-fill**: Country auto-detected from region entered in search bar

2. Guide Download Form (`#guideOverlay`)

- **Fields**: Email, Phone, Name, Company, Property type toggle, Number of units
- **Trigger**: Clicking any region card (currently Andalucia only)
- **Flow**: Form → Success view → Download PDF button

Key Dependencies

Package	Version	Purpose
`express`	^4.x	HTTP server and static files
`@anthropic-ai/sdk`	^0.x	Claude API client
`dotenv`	^16.x	Environment variable loading
`cors`	^2.x	Cross-origin request handling
`pdf-parse`	^1.x	PDF text extraction (for `extract-guides.js`)
CDN: jsPDF	2.5.1	Client-side PDF generation

Adding a New Region

1. Drop the PDF guide into `Guias/`

2. Add the mapping in `scripts/extract-guides.js` (filename → slug, name, aliases)
3. Run `npm run extract` to generate the `.md` file and update `_index.json`
4. Add guide content to `guideData` in `index.html` (both `en` and `es` versions)
5. Add the region card to the HTML grid
6. Add city/province aliases to `detectRegionSlug()` in the frontend and `CITY_TO_REGION` in `server.js`

Future Improvements

Short-term Enhancements

- **Expand PDF guides to all regions**: Currently only Andalucia has the full PDF generation. Replicate the `guideData` structure (EN/ES) for all 9 remaining regions
- **Server-side lead storage**: Currently forms don't persist data. Add a database (PostgreSQL/MongoDB) or webhook to a CRM (HubSpot, Salesforce) to store lead captures
- **Conversation history persistence**: Store chat sessions server-side for analytics and follow-up
- **TF-IDF guide filtering**: When the knowledge base grows beyond 50 guides, implement relevance scoring instead of sending all matching guides to Claude

Medium-term Features

- **Multi-language expansion**: Add Italian, Portuguese, French, and German guide translations (matching the Chekin.com language offering)
- **Interactive compliance checker**: Step-by-step wizard that asks questions and builds a personalized checklist for the user's specific situation
- **Regulation change alerts**: Monitor official government sources for regulation changes and notify affected users
- **Property dashboard**: Let registered users track their compliance status per property
- **Lawyer marketplace**: Match users with verified lawyers by region, with appointment booking and reviews

Technical Improvements

- **Vector search / RAG**: Replace keyword-based guide resolution with embedding-based retrieval for more accurate context selection
- **Caching layer**: Cache frequent Claude responses to reduce API costs and latency
- **Analytics dashboard**: Track question topics, regions, conversion rates, and guide downloads
- **A/B testing**: Test different system prompts, follow-up strategies, and UI layouts
- **Rate limiting upgrade**: Move from in-memory to Redis-based rate limiting for multi-instance deployment
- **Authentication**: Add user accounts for personalized experience and saved conversations
- **API versioning**: Version the `/api/ask` endpoint for backwards compatibility as the API evolves
- **Webhook integrations**: Send lead captures to Slack, email, or CRM automatically
- **PDF generation server-side**: Move PDF generation to the server using Puppeteer or similar for more reliable, consistent output with full font/unicode support
- **Mobile app**: React Native or Flutter wrapper for the compliance assistant